

ics manufacturing and components.

Collectively, the schemes cover the entire value chain—from chip design, fabs and advanced packaging to materials, manufacturing equipment, R&D, start-ups and talent development. “This is a holistic ecosystem,” says Chandak. “Each segment drives growth in another, creating a virtuous cycle that can strengthen India’s entire electronics and semiconductor industry.”

TIME FOR START-UPS

For venture investors, the announcements are much more than another government incentive. They fundamentally change their investable universe in India. Vishesh Rajaram, Founding Partner at Speciale Invest, says earlier ISM was about anchoring large manufacturing capacity. This phase explicitly funds chip design, equipment, materials and packaging, the layers where start-ups, and not just conglomerates, compete.

“UPI didn’t just move payments. It created the rails for hundreds of fintech start-ups to build on. This time, India’s semiconductor policy has been architected with that same ecosystem logic,” says Rajaram.

For founders building chips, the schemes change how they build products. Take Mindgrove Technologies, an IIT Madras-incubated fabless semiconductor start-up developing indigenous processors for applications such as computer vision, security and edge computing. Shashwath T. R., the co-founder and CEO, says the real opportunity for start-ups isn’t owning billion-dollar fabrication plants but building products around them. “Manufacturing is never going to be a start-up game,” he says. “Start-ups can create value in chip design, intellectual property and specialised products.”

CREDIBILITY TO CAPABILITY

Semicon 1.0 was about proving India is a reliable destination, while Semicon 2.0 is looking at creating a self-sustaining, interconnected industrial ecosystem

SEMICON 1.0

▶ OUTLAY OF ₹76,000 CRORE TO DEVELOP A DOMESTIC SEMICONDUCTOR AND DISPLAY MANUFACTURING ECOSYSTEM

FOUR-PRONGED APPROACH COMPRISING

- ① SEMICONDUCTOR FABS
- ② DISPLAY FABS
- ③ COMPOUND SEMICONDUCTORS/SILICON PHOTONICS/SENSORS, DISCRETE SEMICONDUCTOR FABS AND ATMP/OSAT FACILITIES
- ④ DESIGN-LINKED INCENTIVE SCHEME

▶ AIM TO ATTRACT LARGE INVESTMENTS TO ESTABLISH SILICON CMOS-BASED SEMICONDUCTOR WAFER FABRICATION FACILITIES IN INDIA, BUILD A TRUSTED SEMICONDUCTOR VALUE CHAIN AND STRENGTHEN INDIA’S ELECTRONICS MANUFACTURING ECOSYSTEM.

SOURCE INDIA SEMICONDUCTOR MISSION

SEMICON 2.0

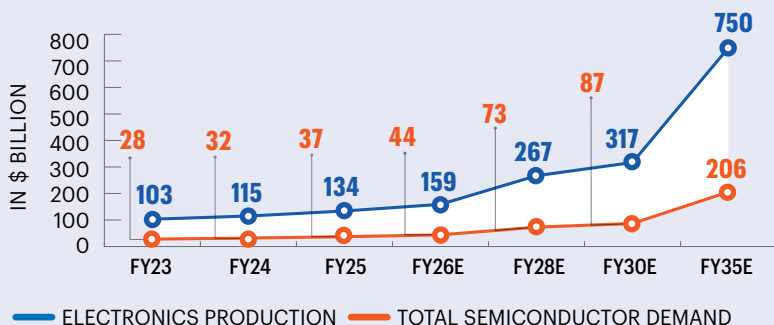
▶ OUTLAY OF ₹1,27,500 CRORE TO BUILD A COMPREHENSIVE SEMICONDUCTOR ECOSYSTEM AND STRENGTHEN INDIA’S POSITION IN THE GLOBAL SEMICONDUCTOR VALUE CHAIN

SIX-PILLAR STRATEGY COVERING THE ENTIRE SEMICONDUCTOR VALUE CHAIN

- ① DESIGN
 - ② EQUIPMENT & MATERIAL
 - ③ SEMICONDUCTOR MANUFACTURING
 - ④ ATMP/OSAT
 - ⑤ R&D
 - ⑥ TALENT DEVELOPMENT
- ▶ FOCUS ON INDIGENOUS IP BY ENCOURAGING FULL-STACK CHIP DESIGN AND COMMERCIALISATION OF INDIAN SEMICONDUCTOR TECHNOLOGIES RATHER THAN ONLY MANUFACTURING.

DEMAND PUSH

India’s semiconductor demand will be driven by rapid growth in electronics manufacturing, expanding data centres and cloud infrastructure, higher chip content in vehicles, especially EVs and ADAS, and AI adoption



NOTE E IS ESTIMATES SOURCE NITI AAYOG